

## Data Flow Diagram Template

|                      |                                                  |
|----------------------|--------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                    |
| <b>Team ID</b>       | Batch 2023-27   ACOE                             |
| <b>Project Name</b>  | Rising Waters – ML-Based Flood Prediction System |
| <b>Maximum Marks</b> | 2 Marks                                          |

### Level 0 — Context Diagram

| Element         | Description                                                                           |
|-----------------|---------------------------------------------------------------------------------------|
| External Entity | Disaster Management Officer / Meteorologist                                           |
| System          | Rising Waters Flood Prediction System (Flask + Random Forest)                         |
| Input Flow      | Temp, Humidity, Cloud Cover, ANNUAL, Jan-Feb, Mar-May, Jun-Sep, Oct-Dec, avgjune, sub |
| Output Flow     | Binary prediction label (Flood / No Flood) + probability score + emergency protocol   |

### Level 1 — Internal Data Flow

| Proc<br>ess<br>ID | Process Name       | Input                | Output                     | Data Store  |
|-------------------|--------------------|----------------------|----------------------------|-------------|
| P1                | Form Input Capture | User web form values | Raw feature dictionary     | —           |
| P2                | Input Validation   | Raw feature dict     | Validated dict / Error msg | —           |
| P3                | Feature Scaling    | Validated features   | Scaled NumPy array         | scaler.save |

| <b>Proc<br/>ess<br/>ID</b> | <b>Process Name</b> | <b>Input</b>        | <b>Output</b>                | <b>Data Store</b>   |
|----------------------------|---------------------|---------------------|------------------------------|---------------------|
| P4                         | ML Prediction       | Scaled NumPy array  | Label (0/1) + Probability    | floods.save         |
| P5                         | Result Routing      | Prediction label    | HTTP redirect to result page | —                   |
| P6                         | Prediction Logging  | All inputs + result | New row appended             | predictions_log.csv |